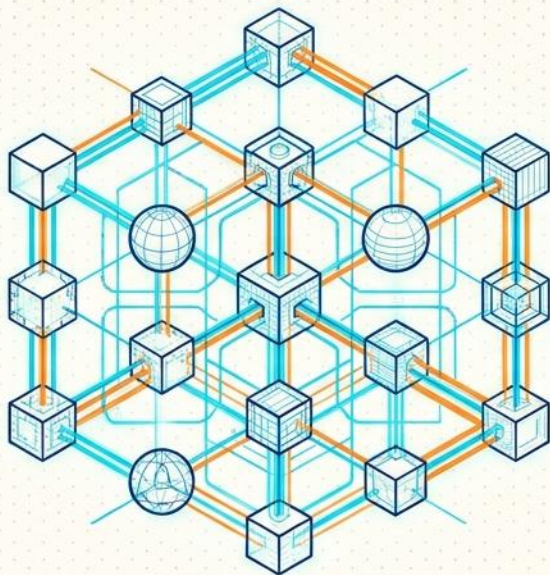




Data Communication & Networks

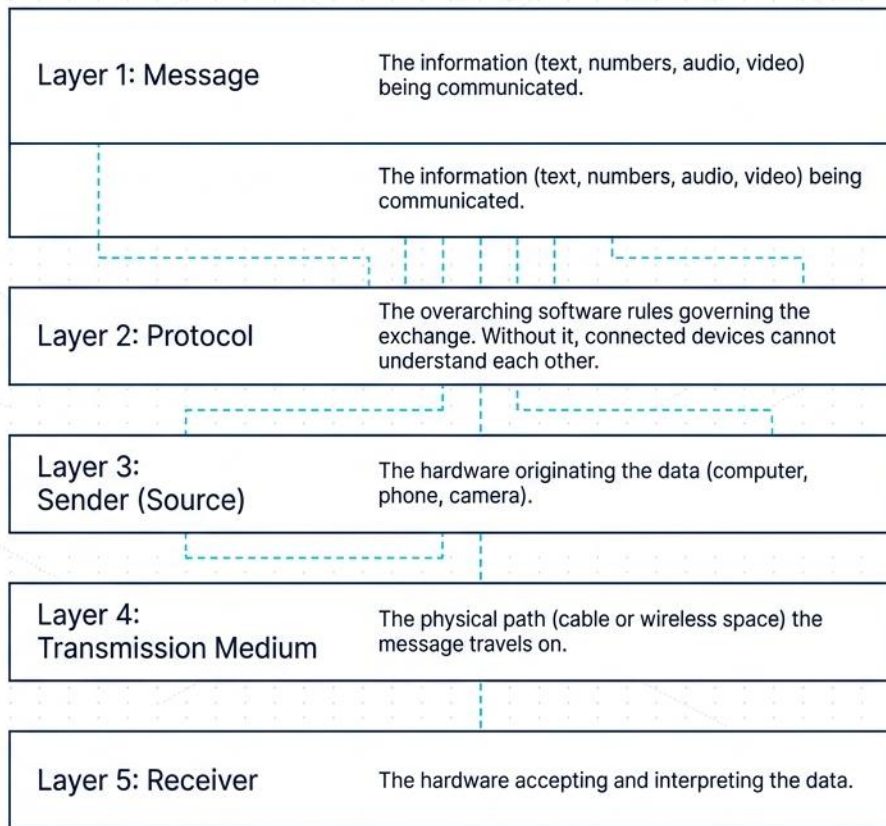
The Architectural Blueprint



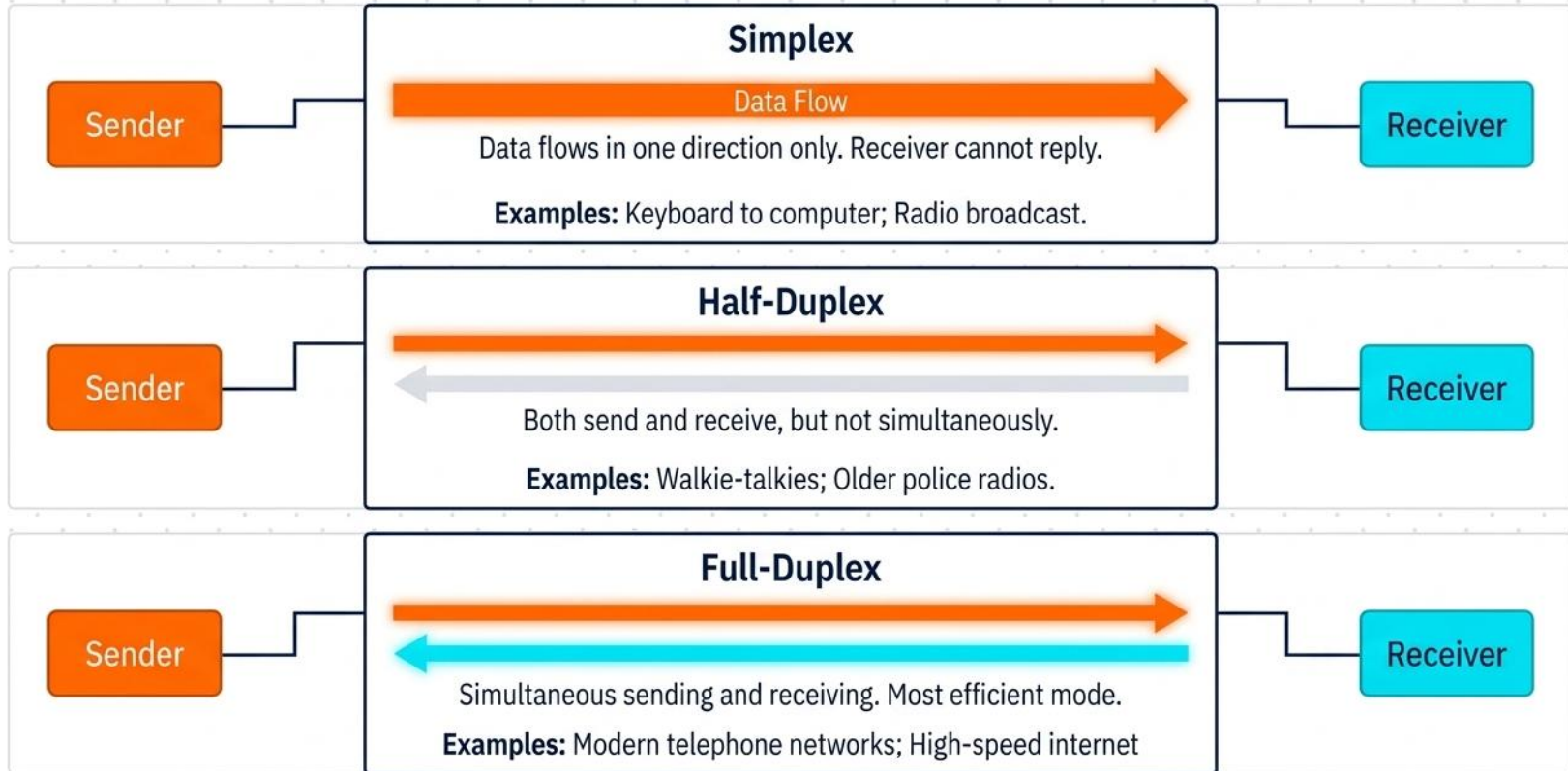
Distilled systems and structures
for Software Engineering

Anatomy of a Communication System

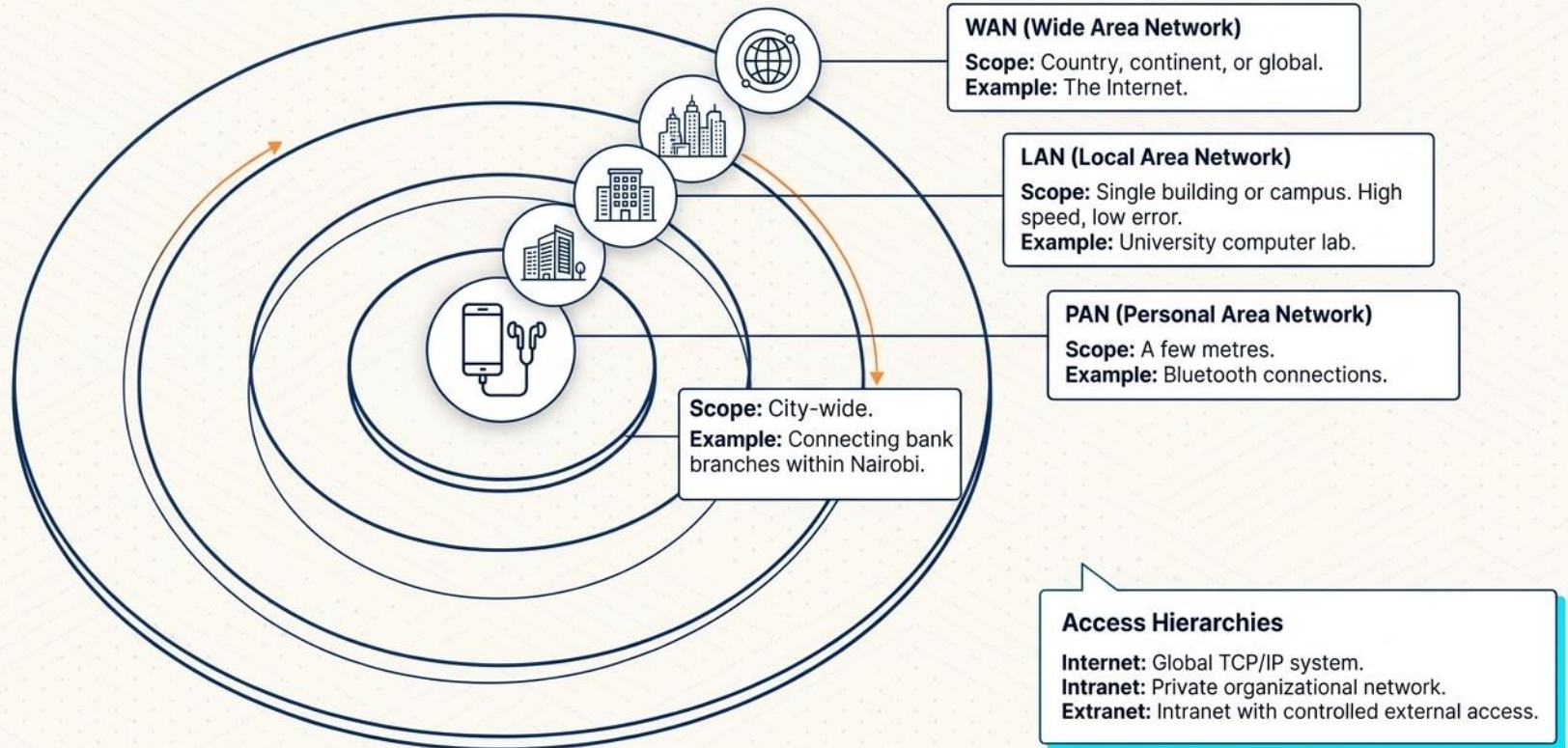
Data communication is the transfer of data from a source to a destination in a form the receiver can understand and use.



Directing the Data Flow



Scaling the Network Architecture

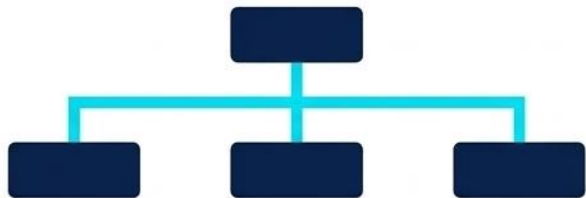


Connection Typologies



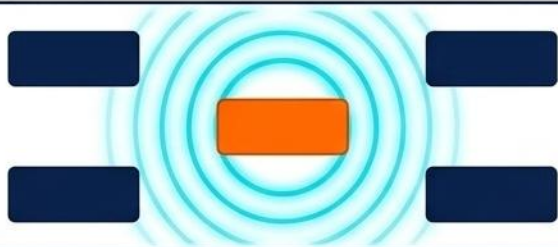
Point-to-Point

Direct dedicated link. Simple and secure, but severely limits scalability.



Multipoint (Multidrop)

Multiple devices share a single link. Cost-effective but requires precise transmission control.



Broadcast

One device sends, all network devices receive simultaneously (e.g., traditional Ethernet hub).

Physical Layouts and Topologies



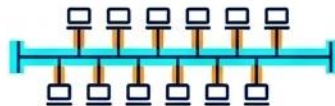
Mesh

Every device connects to every other. Highly reliable for critical systems, but expensive and complex to wire.



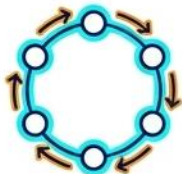
Star

All devices connect to a central switch. Easy to manage, but the hub is a single point of failure.



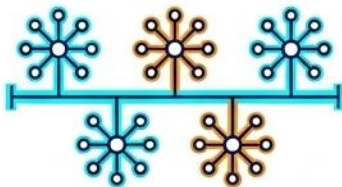
Bus

Devices share one backbone cable. Cheap and simple, but backbone failure kills the network.



Ring

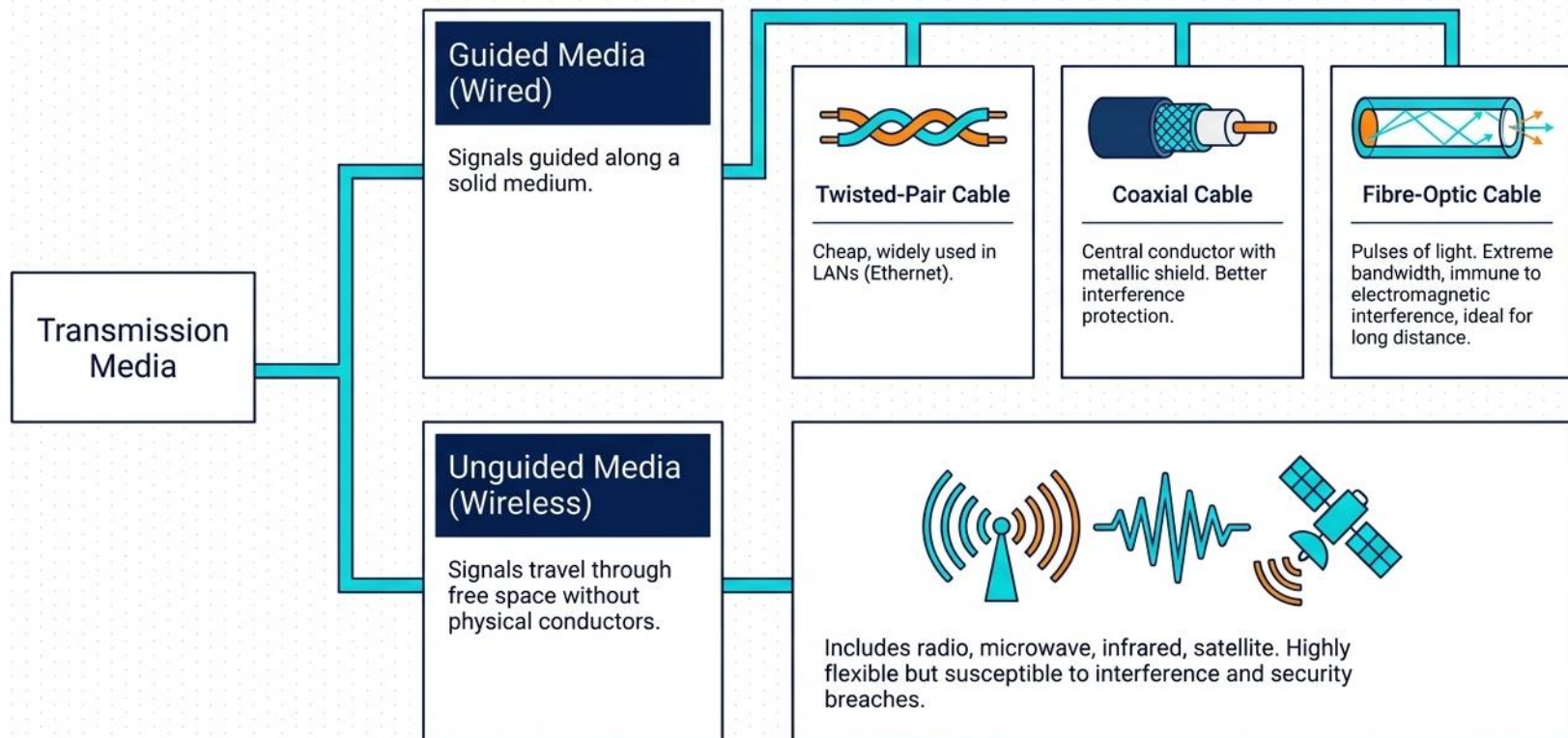
Connects to exactly two neighbors. Unidirectional data. One failure breaks the ring unless dual-loops are used.



Hybrid

Combines multiple topologies. The standard for large modern networks.

Transmission Media Pathways



Navigating the Intersections: Switching Techniques

Circuit Switching

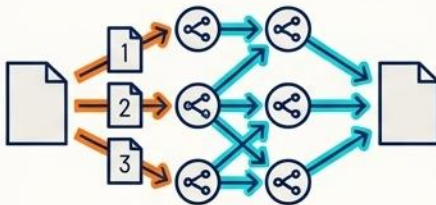


Establishes a dedicated path for the entire conversation. Three phases: Establish -> Transfer -> Terminate.

Drawback

Inefficient; resources remain reserved even during silence (e.g., traditional phones).

Packet Switching

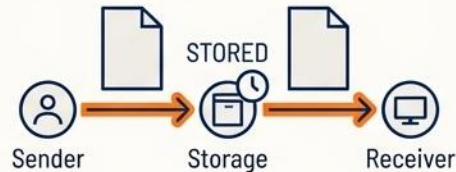


Data broken into independent packets navigating optimal routes. Reassembled at destination.

Advantage

Highly efficient resource use. The backbone of the Internet.

Message Switching



Store-and-forward. Entire message moves node-to-node.

Detail

Less common in modern rapid networking.

Evaluating System Integrity

Three criteria define a successful communication network:

Performance



Measured by Throughput (actual data delivered) and Latency (delay from source to destination).

Dependencies:

Hardware capability, medium type, and user load.

Reliability



Uptime, time-to-recovery, and fault tolerance.

Dependencies:

Resilience against disasters and hardware failure.

Security

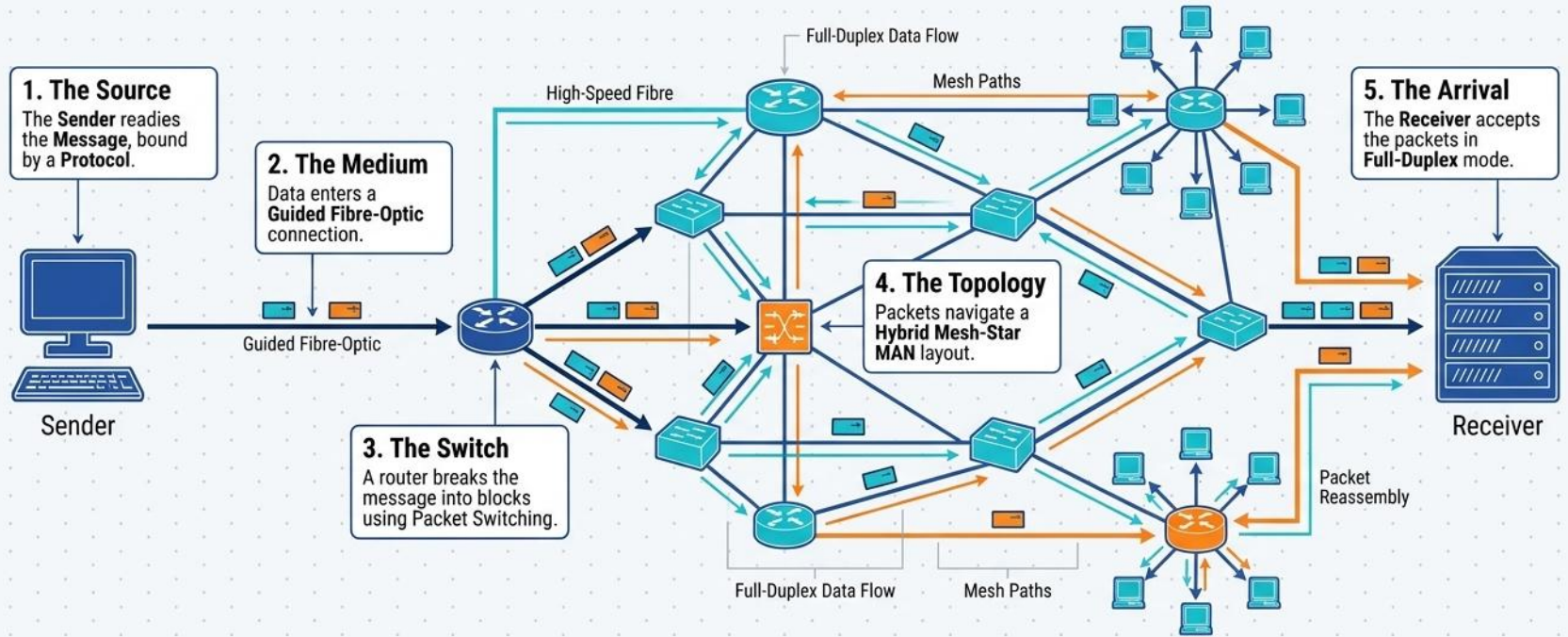


Data protection and integrity.

Dependencies:

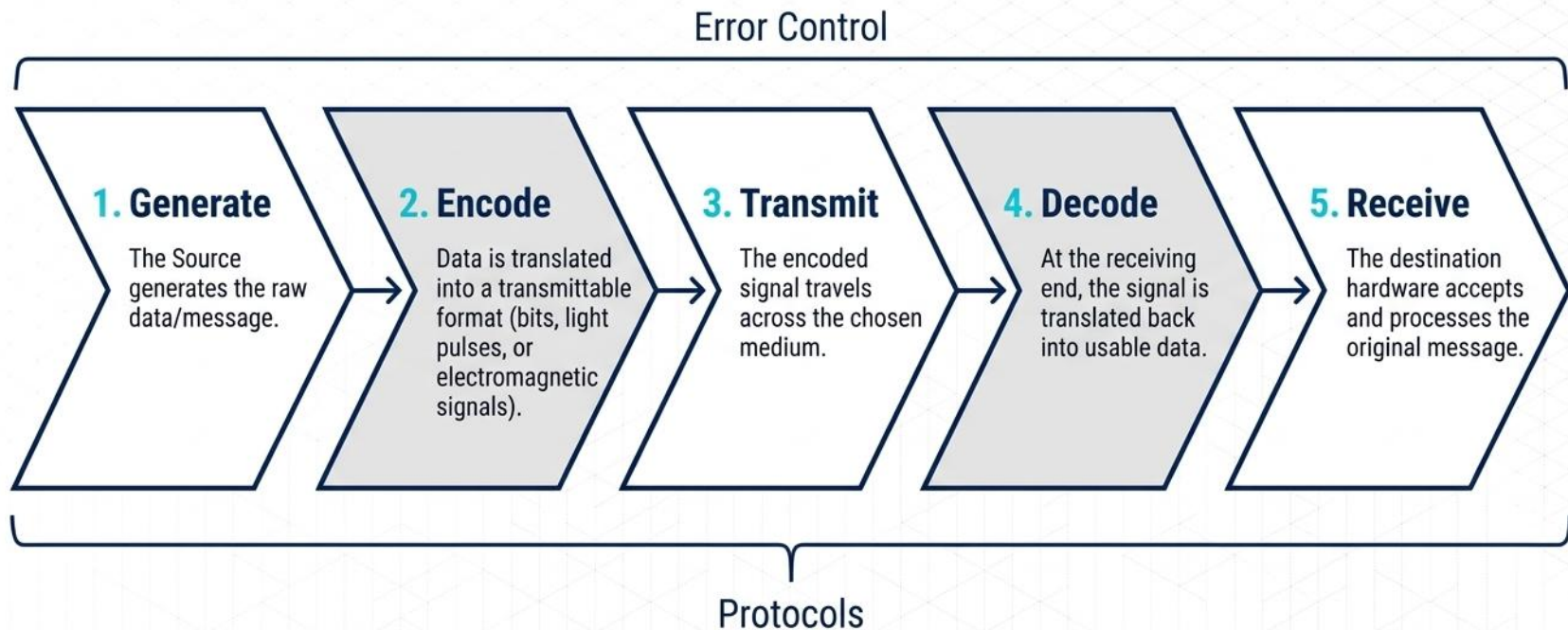
Defenses against unauthorized access, eavesdropping, and cyber attacks.

SYNTHESIS: The System in Motion



All 5 components, transmission modes, topologies, media, and switching techniques execute simultaneously in fractions of a second to deliver a single payload.

The Mechanics of Transmission



System Guardrails and Governance



Flow Control

Prevents sender from overwhelming receiver; prevents buffer overflow and data loss.



Error Detection

Identifies corruption during transmission. Uses Parity bits, Checksums, and CRC.



Addressing

Ensures correct destination delivery. Splits into MAC (Physical) and IP (Logical) addresses.



Routing

The logic of selecting the most optimal path across networks using routing tables.

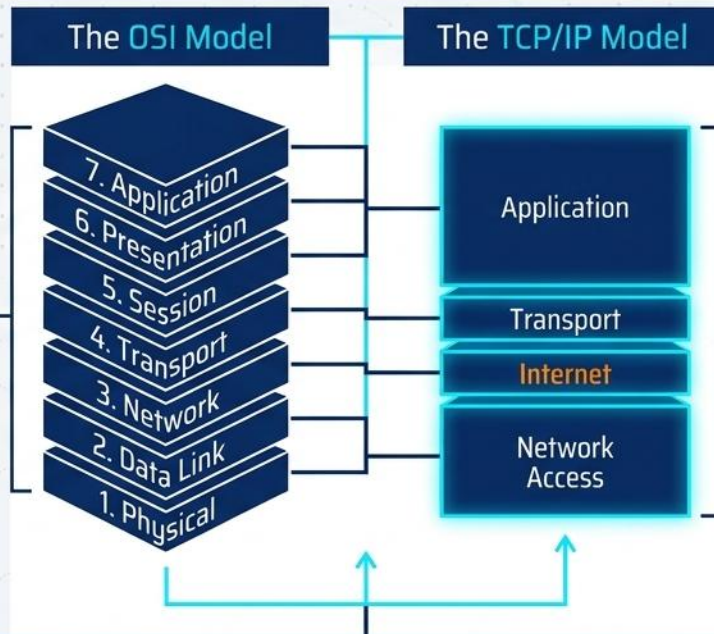


Network Management

The oversight umbrella: Security, bandwidth monitoring, fault recovery, and user accounting.

Standardizing the Architecture: Reference Models

Models provide a **structured language** to define the functions required for successful data communication between fundamentally different systems.



Details

Open Systems Interconnection

A 7-layer theoretical model developed by ISO.

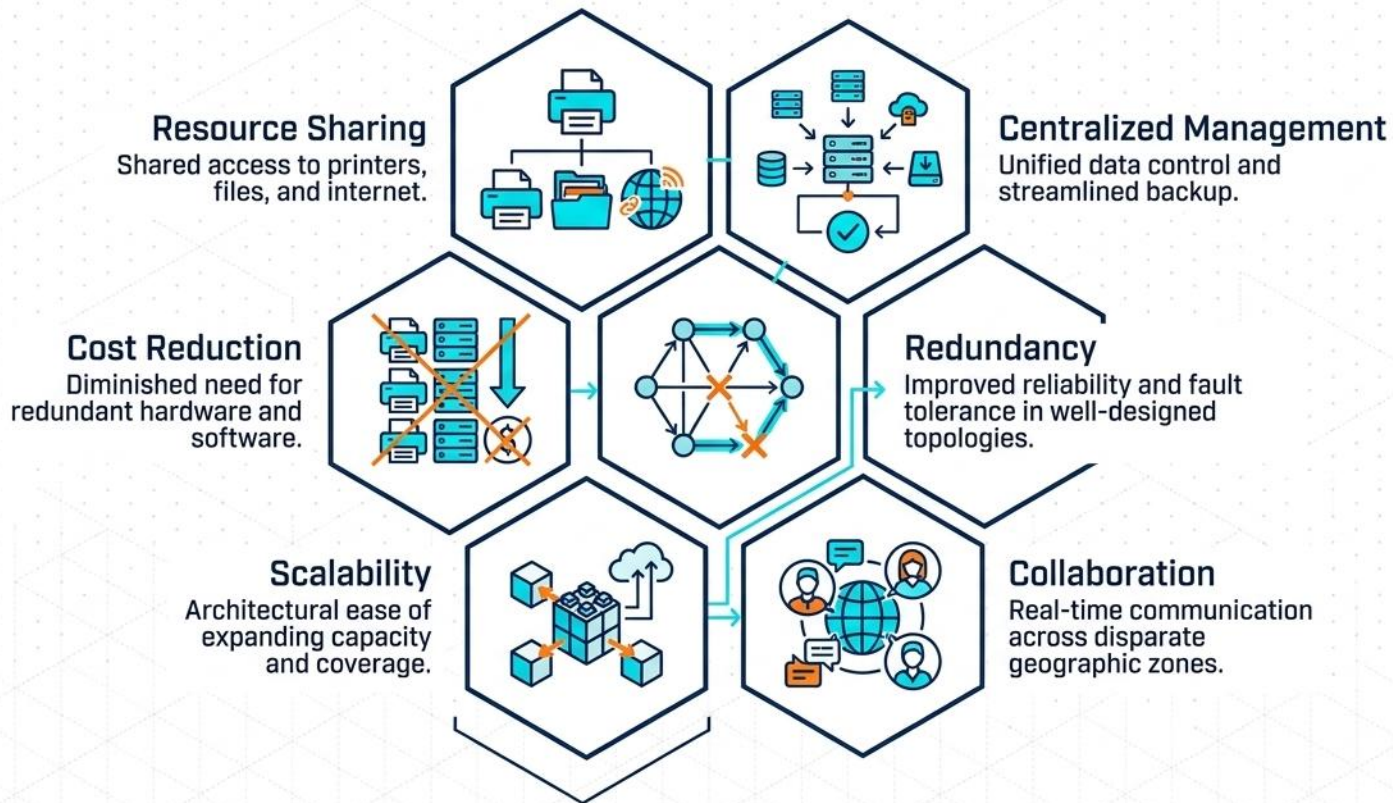
Exceptional for teaching and isolating networking concepts.

Details

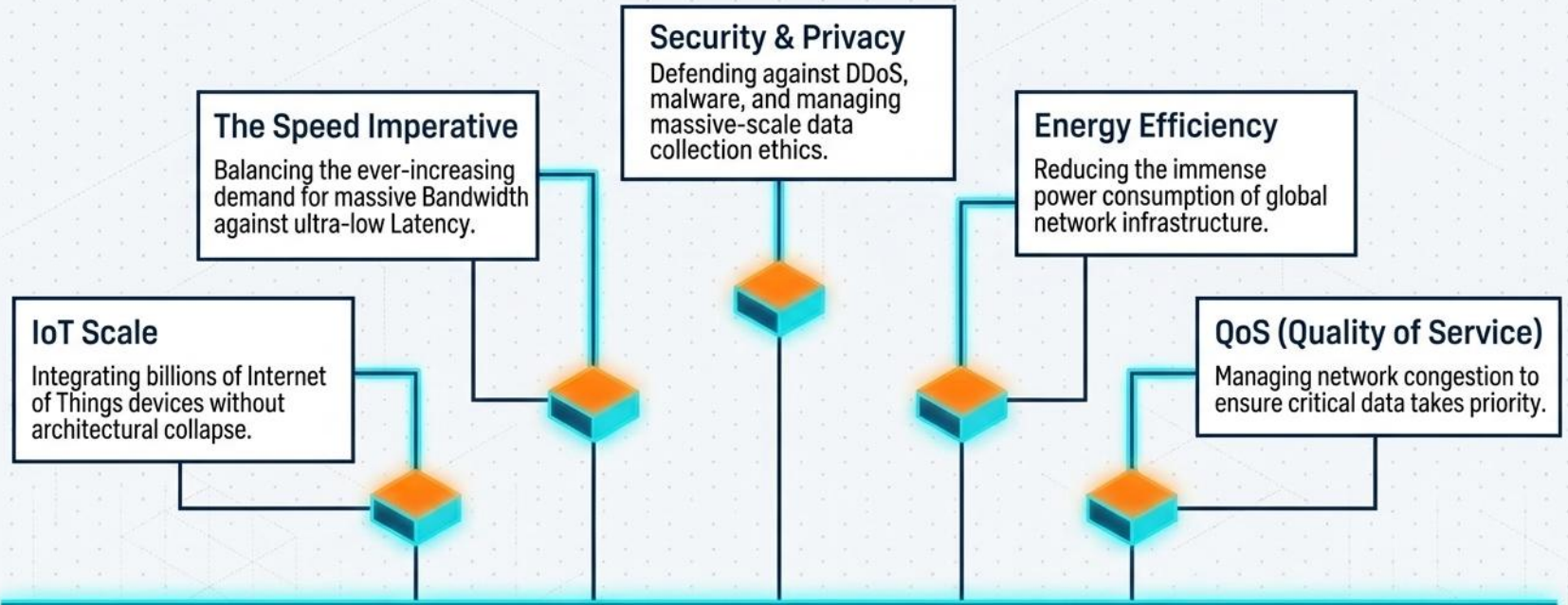
The practical, operational foundation of the modern Internet.

Streamlined for real-world deployment.

The Value Proposition of Connectivity



Emerging Frontiers in Networking



The blueprint is established. The future relies on how we scale it.